MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 8020W

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Diode, half-wave rectifier, high vacuum

See figure 1

Mounting position: Any

Weight: 4-ounces (113.4 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epx	ib	Ib	Ⓒ tk	Pp
Unit:	Vac	kv	ma	mAdc	sec	W
Maximum:	5.25	40	600	100	---	75
Minimum:	4.75	---	---	---	5	---
TEST CONDITIONS:	5.0	---	---	---	5	---

GENERAL:

Qualification - Required

Ⓒ denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1231	Peak emission	eb = 4,000 v	0.65	II	is	1.6	---	a
1301	Filament current		0.65	II	If	5.5	6.5	Aac
1353	Operation of rectifiers	epx = 40 kv; ib = 600 ma; Ib = 100 mAdc	0.65	II	---	---	---	---
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	No voltages applied; t = 120 seconds	---	---	---	---	---	---
1041	Shock	Hammer angle = 25°; no voltages applied	---	---	---	---	---	---
---	Shock-test end point:							
1353	Operation of rectifiers		---	---	---	---	---	---
1256	Electrode current (anode)	Eb = 200 Vdc	---	---	Ib	75	---	mAdc
	Quality conformance inspection, part 3							
---	Life test	Group C; operation of rectifiers; t = 500 hours	---	---	---	---	---	---
---	Life-test end points:							
1231	Peak emission		---	---	is	1.4	---	a
1353	Operation of rectifiers		---	---	---	---	---	---
(C) 1031	Variable frequency vibration	No voltages applied (see note)	---	---	---	---	---	---

(C) NOTE: This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection (QCI), part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

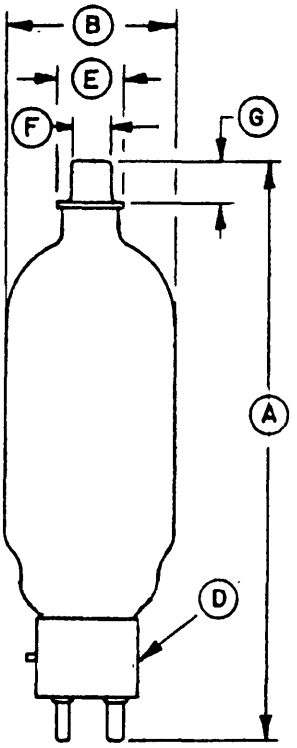
Review activities:
Air Force - 80
DSA - ES

User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

(C) Preparing activity:
Air Force - 85

(Project 5960-2956)

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PIN CONNECTIONS	
Pin No.	Element
1	f
2	nc
3	nc
4	f
Cap	a

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	---	8.000 (203.20)
B	---	2.312 (58.72)
E	.844 (21.44)	.906 (23.01)
© Quality conformance inspection, part 3 (see note)		
D	Base: A4-10 (EIA)	
F	.559 (14.20)	.573 (14.55)
Reference dimensions		
G	.500 (12.70)	

NOTE:
Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 8020W.

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OMB Approval
No. 22-R255

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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